Unlimited Number of Spanning Clusters in 3D Percolation Above the Critical Point

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Inspired by the formation of bigels, we developed a bond percolation model that yields multiple percolating clusters in three dimensions not only at the critical point, but also above it. Our simulations suggest that, in the thermodynamic limit, there is no upper limit to the number of percolating clusters. In finite systems the maximum number of percolating clusters that can be obtained grows logarithmically with the lattice size. For equal initial densities of species in the thermodynamic limit, all clusters percolate at the same threshold and exhibit critical exponents consistent with the critical exponents of classical percolation. The threshold depends linearly on the initial density of species and the maximum and minimum initial densities decay exponentially with the maximal number of spanning clusters. For the scenario of species with different initial densities, the percolation thresholds are not the same.

I. INTRODUCTION

Percolation, the paradigm for random connectivity, has since Broadbent and Hammersley [1] become one of the most widely applied statistical models [2, 3]. It has been used successfully to describe a wide range of phenomena in nature, including the sol-gel transition [4], incipient flow through porous media [5, 6], epidemics [7], and network failure [8].

In a recent development [9, 10], it was shown that a class of materials called "bigels", which form through arrested demixing in binary colloidal mixtures may develop two simultaneously spanning clusters, one for each species. Other studies have also reported experimental realizations of bi-percolated structures [11, 12] and tri-percolated structures [13, 14].

The emergence of multiple percolating clusters in these systems can happen only because aggregates of different species cannot merge. In fact, it has been proven [15, 16] that for percolation with no restriction on the merging of clusters, the number of infinite clusters is either zero, one or infinity.

However, at the percolation threshold p_c one must distinguish between infinite and spanning clusters [17–19]. In the case of spanning clusters, for $d > 1$ the probability of finding more than one spanning cluster at the critical point p_c does not go to zero as $L \to \infty$ [17]. In fact, for $d = 2$, an exact result for the probability P_k of having k spanning clusters in two dimensions was derived using conformal invariance [20]. In another study [21], the go-with-the-winners strategy [22] was applied to a rectangular lattice with periodic boundaries in one direction to simulate five simultaneously spanning clusters [23] and calculate the probability of finding them.

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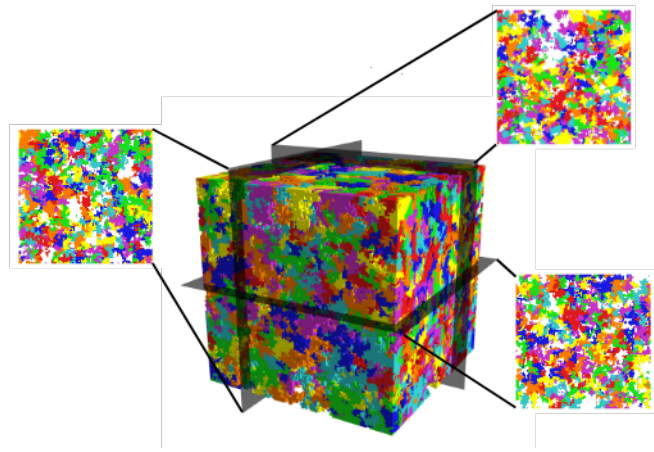

FIG. 1. Percolating clusters of seven species in a simulation on a simple cubic lattice with side length $L = 128$. Each color corresponds to a different species. The empty regions between the percolating clusters correspond to clusters that do not percolate by the end of the process. The black planes indicate the cuts of the configuration shown.

To achieve more than one percolating cluster above the critical point, the polychromatic model was introduced [24], where nodes or bonds are differentiated by species, enabling n percolating clusters, though constrained by n times the percolation density at the critical point being less than unity. This concept was expanded upon models with species-dependent bond formation [25–27], correlated two-component percolation [28], site percolation with independent concentrations [29], AB percolation [30, 31], colored percolation [32], two-color percolation [33], polychromatic random networks [34], and tricolor self-avoiding walks [35]. However, none of these models can obtain more than $1/p_c$ percolating clusters.

In a recent paper [36], we presented a model that allows for the percolation of $k > 1/p_c$ spanning clusters not only

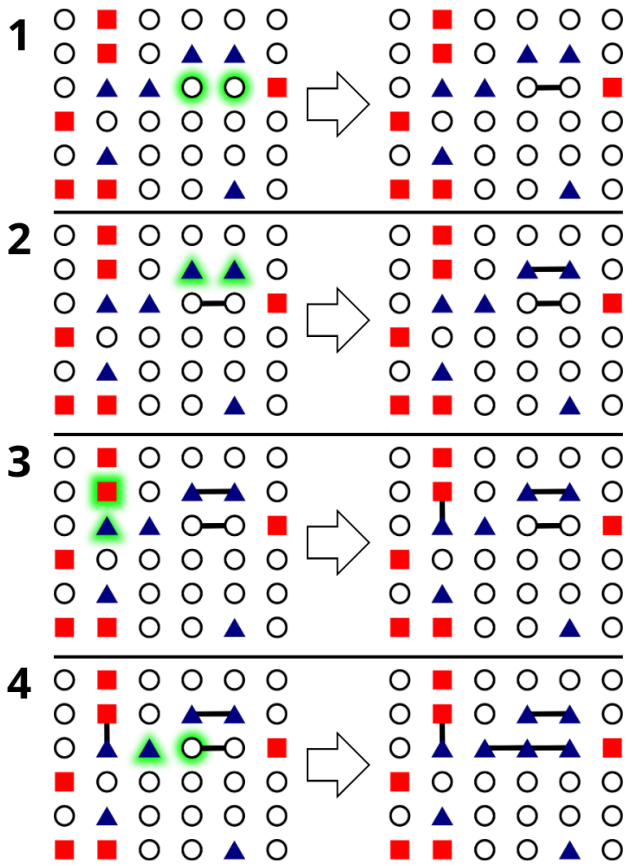

FIG. 2. This figure illustrates the rules of our model. The red squares and blue triangles represent sites inhabited by two different species, while the white circles denote uninhabited sites. On the left, the system's configuration is shown before the bond is connecting the sites highlighted in green are occupied, and on the right, the system is displayed after the bond has been occupied. In Case 1, a bond connects two uninhabited sites. In Case 2, a bond connects two sites inhabited by the same species. In Case 3, a bond connects sites inhabited by different species, but both retain their species identities. In Case 4, when a bond connects an uninhabited node to a node inhabited by a certain species, all sites in the inhabited cluster will be inhabited by the species of the inhabited node.

at the critical point but also above it on 3D lattices, as illustrated in Fig. 1, which shows the final configuration of seven percolating clusters. The main purpose of the present work is to extend the results presented in that paper. We analyzed three types of lattices: the simple cubic lattice, the diamond lattice [37], and the simple cubic lattice with additional bonds between nearest and second nearest neighbors (2NN) [38]. We seek to identify the necessary conditions to obtain k spanning clusters and determine whether there is an upper limit to the number of percolating clusters. Additionally, we analyze the structure of these clusters. Our results suggest that it may be experimentally feasible to create systems exhibiting multiple percolating clusters.

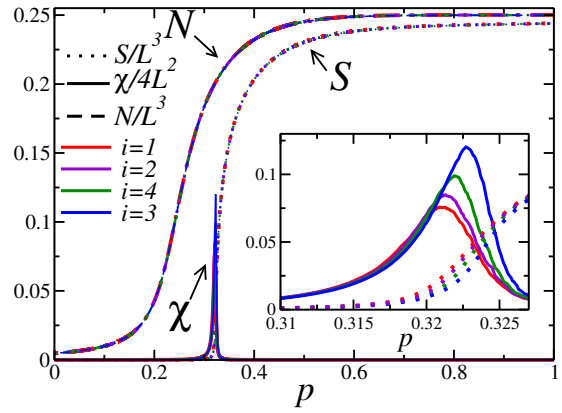

FIG. 3. Evolution of properties of the cluster distribution during the bond occupation process. Species are labeled in the order in which they percolate, that is, we label by 1 the species that first percolates, by 2 the second species to percolate and so on. The curves are averages over 1000 samples. The colors of the curves indicate the order i in which the species percolated. The dotted lines correspond to the average size S of the largest cluster; the solid lines indicate the second moment of the cluster size distribution χ, as defined in Eq. (1); the dashed lines indicate the number of nodes N that have been occupied by each species. For convenience, each of the quantities was rescaled by the system size in a different way, as indicated in the legend. We used a simple cubic lattice with size $L = 512$, 4 species, and initial density $\rho = 0.0047$. The inset shows that the curves for the second moment χ and the size of the largest cluster S present some dependence on the percolation order i.

The paper is organized as follows: In Sec. II, we introduce the model. In Sec. III, we discuss the critical properties of the model, including the second moment of the cluster size distribution, the largest cluster, and the critical point. In Sec. IV, we analyze the topological properties, such as fractal dimension, shortest path, and interfacial contact between the largest clusters at the critical point. In Sec. V, we study the minimum and maximum limits of the initial density. In Sec. VI, we show how to determine the minimum side length required to find k percolating clusters. In Sec. VII, we examine the scenario where two species have different initial densities. Finally, in Sec. VIII, we present the conclusions.

II. MODEL

We consider a simple cubic lattice of size L, where initially all bonds are empty ($p = 0$) and a fraction $k\rho \leq 1$ of sites is randomly and permanently inhabited by k different species. Precisely, at the beginning, each species inhabits an equal fraction ρ of the sites

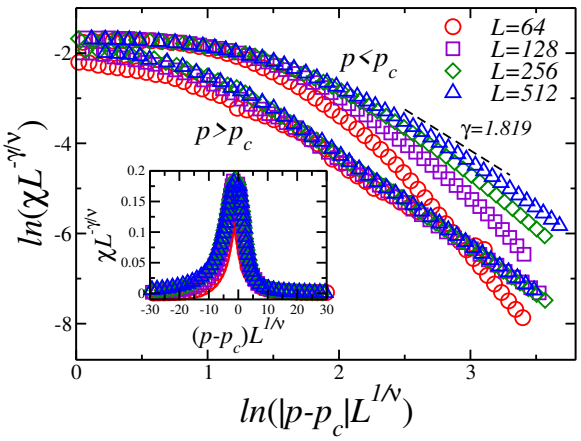

FIG. 4. Logarithmic plot of the finite-size scaling of χ for the first percolating cluster. We used the exponents $\gamma = 1.819 \pm 0.003$ [39], and $\nu = 0.875 \pm 0.001$ [40] of standard 3D percolation. The percolation threshold was $p_c = 0.321$. The slope of the dashed line indicates the known value of γ. The averages of this figure were done over 1000 samples on a simple cubic lattice for sizes L shown in the legend and we considered 4 species and $\rho = 0.0047$. The inset shows the same data on a linear scale.

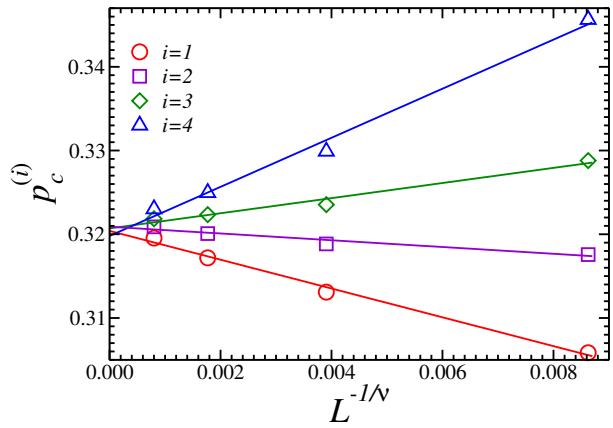

FIG. 5. Percolation threshold p_c as a function of $L^{-1/\nu}$ for four different species. For each realization, we record $p_c^{(i)}$ as the fraction p of occupied bonds when the ith percolation transition takes place. In case more than one cluster of same species connects the boundaries, we record the first one percolating. The thresholds of $p_c^{(i)}$ are the averages of these percolation thresholds over 1000 realizations, for $L = 64$, and over 100 realizations for $L = 128$, 256, and 512 for simple cubic lattices. Here we considered the initial density $\rho = 0.0047$, for which we always find four percolating species for all investigated system sizes. The curves follow a function given by Eq. (3) and the extrapolation of the straight lines suggests that in the thermodynamic limit, $L \to \infty$, all the clusters percolate at the same fraction $p_c = 0.321 \pm 0.001$. The statistical errors are smaller than the size of the symbols and were therefore omitted them from the figure. Here, we used the exponent $\nu = 0.875 \pm 0.001$ of standard 3D percolation [40].

of the lattice, with the constraint that no site can be inhabited by more than one species. Note that here, we use the terms "empty" and "occupied" for bonds and "inhabited" and "uninhabited" for sites. Then, as in bond percolation [41], one starts to randomly select bonds and to occupy them, while ensuring that every bond can only be chosen once. When a chosen bond connects a site inhabited by a given species with an uninhabited site, then this site and all the uninhabited sites that may belong to its cluster will become inhabited by the same species. A cluster is defined here as a group of connected sites that are either uninhabited or inhabited by one single species. This rule allows the spreading of a species into uninhabited territory through the aggregation of clusters. A final and crucial rule is, that if two neighboring sites are inhabited by different species, they retain their species identities even if they are connected by an occupied bond, therefore belonging to different clusters. Fig. 2 illustrates the rules of the model. The case $k = 1$ is equivalent to classical percolation, while $\rho k = 1$ corresponds to the polychromatic model.

We implement periodic boundary conditions in directions x and y and open boundaries in z direction. We define n_b as the number of bonds that have been occupied. Therefore the probability of a bond occupation is $p = n_b/(3L^3 - L^2)$ which will be the variable controlling the percolation process. The properties of the model will be investigated as a function of two parameters, namely, the number of different species k and the density ρ of nodes initially inhabited by each species, and, except for Sec. VII, only the case where the initial density ρ is the same for all species will be considered.

We search for percolating clusters that span from the top to the bottom of the 3D lattice along the z axis. Growing clusters of different species act as obstacles to one another. In this way, each species races to first reach the uninhabited nodes while blocking the growth of the competing clusters of other species.

In some cases, it is also possible to observe more than one percolating cluster of the same species. For example, for $\rho = 0.025$, $L = 512$ and $k = 4$, we observe with a probability of approximately 0.47 at least two percolating clusters of the same species at the critical point. Most of these spanning clusters of same species will merge to a single cluster with increasing p before the end of the occupation processes. However, in some cases, it is even possible to observe two percolating clusters of the same species at $p = 1$. For instance, for $\rho = 3.89 \times 10^{-4}$, $L = 512$ and $k = 9$, this happens for about 2% of the samples.

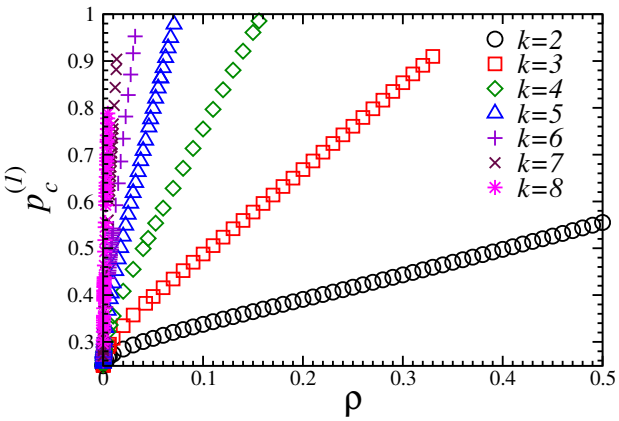

FIG. 6. Critical point of the first $p_c^{(1)}$ percolating cluster as a function of the initial density ρ. Each color represents a different number k of species. We observe that the average percolation threshold grows linearly with ρ as in Eq. (4) and exponentially with k as in Eq. (5). The value of ρ at which $p_c^{(1)} = 1$ is the maximum initial density ρ_{max} above which no percolating cluster should be observed in the thermodynamic limit. More details on this topic are discussed in Sec. V. The averages were performed over 100 samples on a simple cubic lattice of size $L = 512$.

III. CRITICAL PROPERTIES OF THE MODEL

In this section, we study on a simple cubic lattice the second moment of the cluster size distribution χ_i, the size of the largest cluster S_i, the total number of nodes N_i, and the percolation threshold $p_c^{(i)}$ for each species i. We found that they exhibit the same critical exponents γ and ν as the ones of standard 3D percolation.

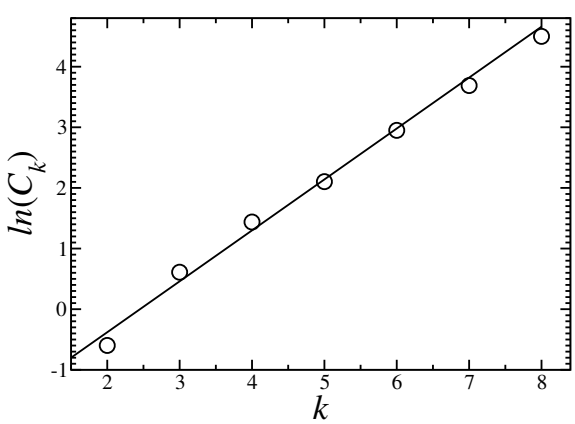

FIG. 7. Semi-logarithmic plot of the slope C_k as a function of k for the first percolating cluster. C_k grows exponentially as a function of k according to Eq. (5). The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

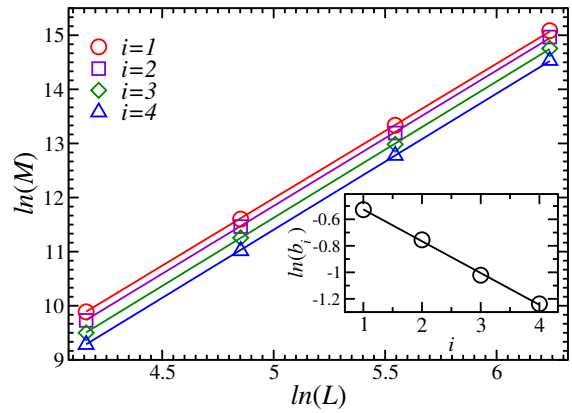

FIG. 8. Logarithmic plot of the mass M of the first (black circles), second (red squares), third (green diamonds), and fourth (blue triangles) percolating clusters at p_c as a function of the side length of the simple cubic lattice L. We consider four species, $\rho = 0.0047$ for each species, and we averaged over 1000 realizations for all sizes L. The fractal dimensions are obtained through the least-squares fit to a power law, $M_i = b_i L^{d_f}$, yielding the exponents $d_f^{(1)} = 2.50 \pm 0.03$ for the first, $d_f^{(2)} = 2.52 \pm 0.02$ for the second, $d_f^{(3)} = 2.54 \pm 0.02$ for the third, and $d_f^{(4)} = 2.53 \pm 0.02$ fourth percolating cluster, respectively. Within error bars, our results suggest that the fractal dimensions of the percolating clusters remain the same, regardless of the percolation order and are consistent with the fractal dimension of 3D standard percolation($d_f = 2.5227$ [42, 43]). In the inset we observe in a semi-logarithmic plot that the prefactors b_i decrease with the percolation order i as in Eq. (7). The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

A. Second moment and largest cluster

We consider S_i as the average number of nodes in the largest cluster of the ith species to percolate, the total number N_i of nodes inhabited by this species, and the second moment χ_i of the cluster-size distribution that is defined here as [2]

$$\chi_i = \frac{\sum_l s_{li}^2 - S_i^2}{N_i}, \tag{1}$$

where s_{li} is the mass of a cluster l inhabited by the ith species. Figure 3 illustrates the evolution of the cluster properties and the inset shows that the curves for the second moment χ and the size of the largest cluster S present some dependence on the percolation order i near the critical point.

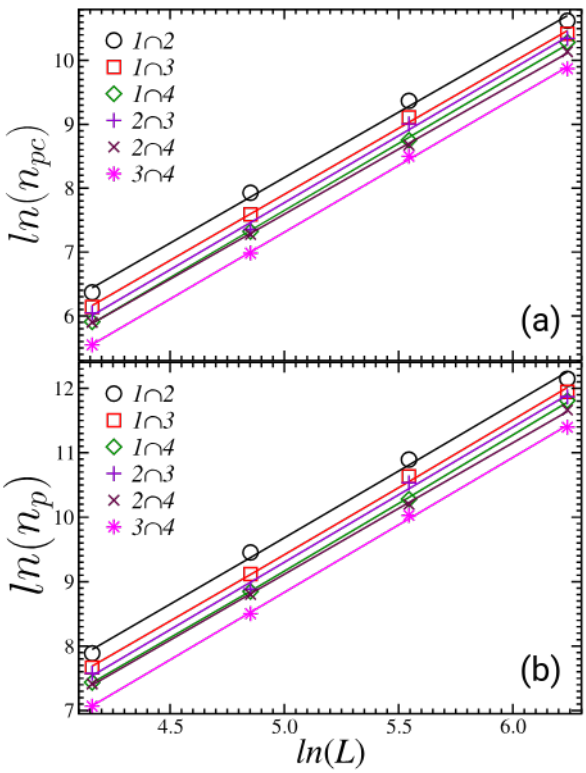

FIG. 9. Logarithmic plot of (a) the number of occupied bonds between the largest clusters, n_{pc}, at p_c, and (b) n_p, the number of bonds between the largest clusters at p_c regardless of whether the bonds had been occupied at p_c, or not, both as functions of the system size L. We consider four species, $\rho = 0.025$, we averaged over 1000 realizations on simple cubic lattices. The different symbols of the curves represent the different values of n_{pc} and n_p between two largest clusters i and j. The exponents were obtained by fitting n_{pc} and n_p at p_c to the power law $n \propto L^{d_f^{\cap}}$. For n_{pc}, the exponents are: $d_f^{1\cap 2} = 2.05 \pm 0.04$, $d_f^{1\cap 3} = 2.07 \pm 0.04$, $d_f^{1\cap 4} = 2.09 \pm 0.05$, $d_f^{2\cap 3} = 2.09 \pm 0.06$, $d_f^{2\cap 4} = 2.04 \pm 0.02$, $d_f^{3\cap 4} = 2.09 \pm 0.06$. Similarly, for n_p, the exponents are: $d_f^{1\cap 2} = 2.06 \pm 0.04$, $d_f^{1\cap 3} = 2.07 \pm 0.04$, $d_f^{1\cap 4} = 2.09 \pm 0.05$, $d_f^{2\cap 3} = 2.09 \pm 0.06$, $d_f^{2\cap 4} = 2.04 \pm 0.02$, $d_f^{3\cap 4} = 2.09 \pm 0.06$. Within error bars, our results suggest that the exponents are consistent with the fractal dimension of the intersection of two fractal objects as in Eq. (8). The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

In the inset of Fig. 4, we show that the curves of the second moment χ for different L collapse on top of each other when the data is rescaled using the critical exponents $\gamma = 1.819 \pm 0.003$ [39] and $\nu = 0.875 \pm 0.001$ [40] of standard 3D percolation. Additionally, we can use this analysis to determine the critical point p_c. Moreover, we know that χ follows the equation [44],

$$\chi(p) \propto |p - p_c|^{-\gamma}, \tag{2}$$

so Fig. 4 illustrates that on a logarithmic scale, the curves display a critical exponent of $\gamma = 1.819 \pm 0.003$ [39].

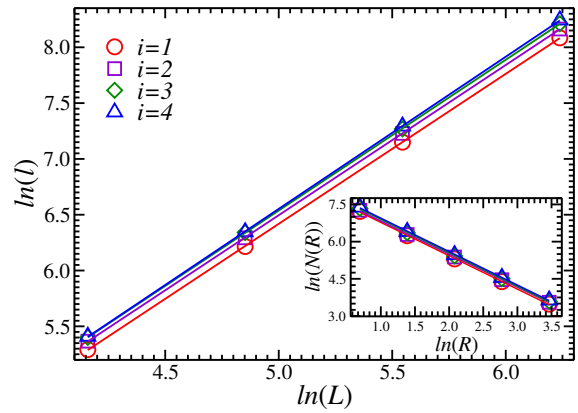

FIG. 10. Logarithmic plot of the shortest path l as a function of L. We calculate l as the number of sites in the shortest path from the top to the bottom of the clusters that has just percolated at p_c. We considered four species, $\rho = 0.025$ for each species, and we averaged over 10000 realizations for all sizes L on simple cubic lattices. The fractal dimensions are obtained through the least-squares fit of the shortest path of the largest cluster of each species to a power law, $l \propto L^{d_{min}}$, yielding the exponents $d_{min}^{(1)} = 1.35 \pm 0.03$ for the first, $d_{min}^{(2)} = 1.35 \pm 0.03$ for the second, $d_{min}^{(3)} = 1.36 \pm 0.02$ for the third, and $d_{min}^{(4)} = 1.36 \pm 0.02$ for the fourth percolating clusters, respectively. The inset shows, on a logarithmic scale, the results of the yardstick method for the shortest paths of the percolating clusters of the cubic lattice with $L = 512$. On the vertical axis, we see the number of covering spheres of radius $N(R)$, and on the horizontal axis, the radius R. The least-squares fit of the shortest path of the largest cluster of each species at p_c to a power law, $N(R) \propto R^{-d_{min}}$, gives the exponents $d_{min}^{(1)} = 1.39 \pm 0.02$ for the first, $d_{min}^{(2)} = 1.39 \pm 0.02$ for the second, $d_{min}^{(3)} = 1.39 \pm 0.02$ for the third, and $d_{min}^{(4)} = 1.39 \pm 0.02$ for the fourth percolating clusters, respectively. Our results suggest that the fractal dimensions of the shortest path are equal within error bars, regardless of the percolation order and are consistent with the fractal dimension of 3D standard percolation $d_{min} = 1.3756 \pm 0.0006$ [45]. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

B. Critical point

Figure 5 presents the average percolation thresholds $p_c^{(i)}(L)$ at which the ith species exhibits for the first time a cluster that reaches both the top and the bottom faces of the system. We observe that the percolation thresholds $p_c^{(i)}(L)$ converge to the same point as the lattice size increases. The least-squares fit to the data is consistent

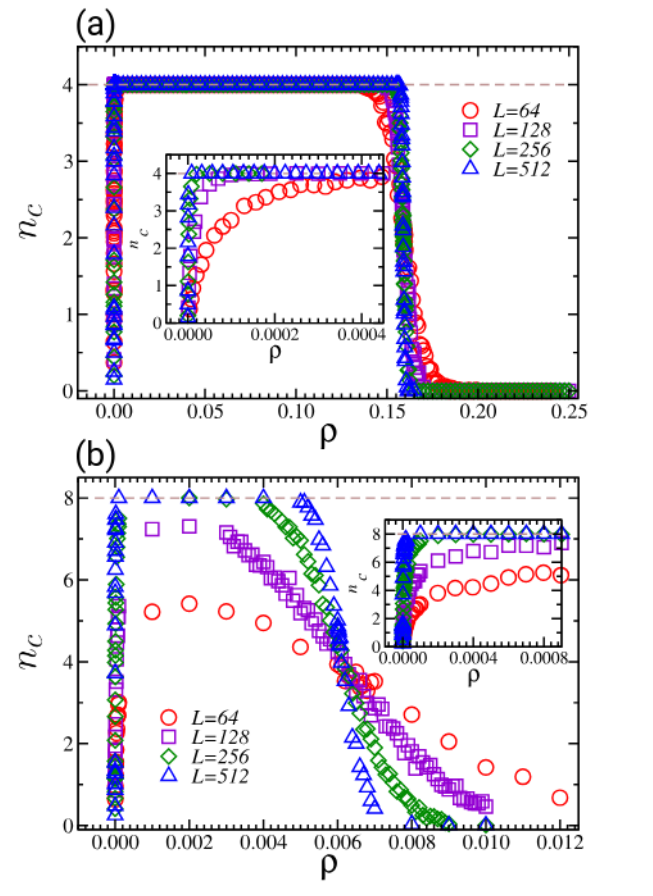

FIG. 11. Average number of percolating clusters of different species n_c as a function of the initial density of inhabited nodes ρ for (a) 4 and (b) 8 species. The legend indicates the side length L of the simple cubic lattice. To guide the eye, we included a horizontal dashed line at the maximum possible number of percolating species. It is noted that for all investigated sizes, ρ must fall within a specific range, $\rho_{min} < \rho < \rho_{max}$, in order to attain the maximum number of percolating species. For $k = 8$, only lattices with $L \geq 256$ reach the maximum number of percolating species. The insets show that in the region around ρ_{min}, n_c depends on the system size L.

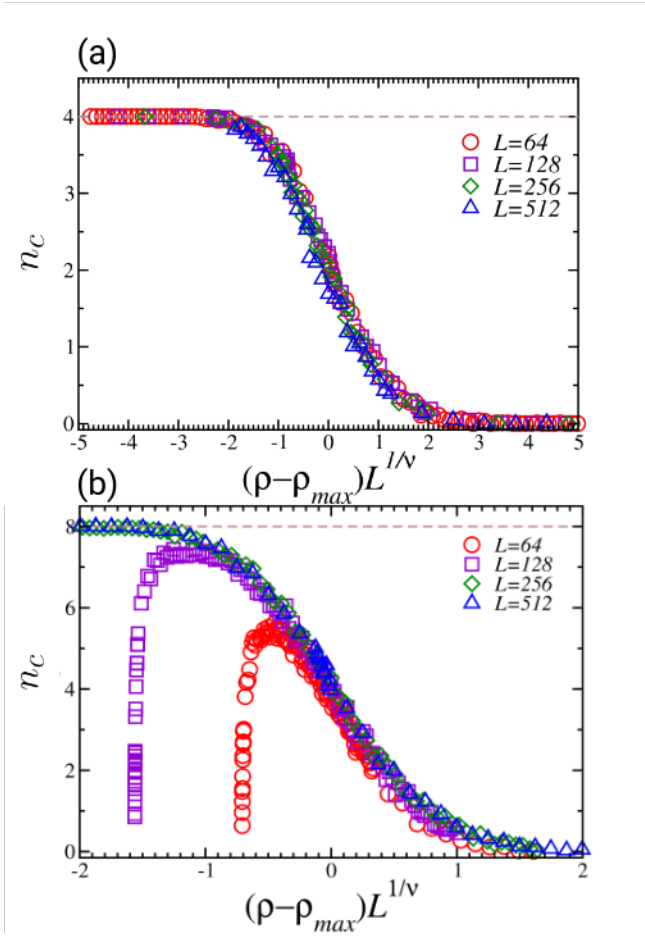

FIG. 12. Finite-size scaling of the average number of percolating clusters n_c as a function of the initial density of inhabited nodes ρ for (a) 4 and (b) 8 species in the region around ρ_{max}. The legend indicates the side length L of each simple cubic lattice. We use a scaling exponent $\nu = 0.875 \pm 0.001$ [40] and $\rho_{max} = 0.159 \pm 0.001$ for (a) and $\rho_{max} = 0.0061 \pm 0.0001$ for (b).

with the following equation [44],

$$p_c^{(i)}(L) = p_c\Big(1 - a_i L^{-\frac{1}{\nu}}\Big), \tag{3}$$

where a_i takes different values for different i and extrapolating L to $L \to \infty$ gives $p_c = 0.321 \pm 0.001$, which is consistent with the scaling shown in Fig. 4. Note that the p_c obtained is much larger than the threshold of standard percolation on the simple cubic lattice, which is $p_c = 0.2488126 \pm 0.0000005$ [40].

The critical threshold p_c depends on the initial density ρ and the number of species k. In Fig. 6, we see, for the first percolating cluster, p_c as a function of ρ for different numbers of species k. We observe that the average percolation threshold grows linearly with ρ as follows,

$$p_c(\rho) = C_0^k + C_k \rho, \tag{4}$$

where the constant C_0^k and the slope C_k are associated with the number of species k. Observe that in Fig.6, as $\rho \to 0$, p_c approaches the threshold of standard bond percolation. Besides that, Fig. 7 shows that the slope C_k follows an exponential law,

$$C_k = Ce^{\alpha k}, \tag{5}$$

where the least-squares fit to the data using this exponential law gives $C = 0.14 \pm 0.02$ and $\alpha = 0.82 \pm 0.03$.

The maximum percolation threshold, $p_c = 1$, corresponds to the maximum initial density, ρ_{max}, above which no percolating clusters can be obtained in the thermodynamic limit. More details on this topic are

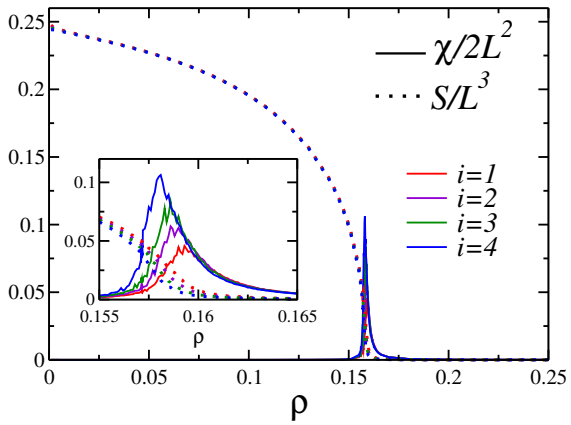

FIG. 13. Evolution of properties of the cluster distribution at $p = 1$ as a function of ρ. Unlike the other figures, species are labeled according to the size of the largest cluster, that is, we label by 1 the species with the largest size, by 2 the species with the second largest size and so on. The curves are the average over 1000 samples. The colors of the curves indicate the size order i. The dotted lines correspond to the average size S of the largest cluster and the solid lines indicate the second moment of the cluster size distribution χ, as defined in Eq. (1). For convenience each of the quantities was rescaled by the system size in a different way, as indicated in the legend. We used a system size $L = 512$ on a simple cubic lattice, with 4 species, and initial density $\rho = 0.025$. The inset shows that the curves for the second moment χ and the size of the largest cluster S exhibit some dependence on the percolation order i.

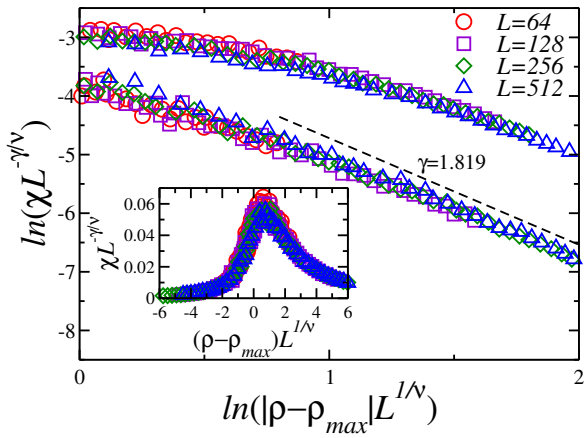

FIG. 14. Logarithmic plot of the finite-size scaling of χ for the first percolating cluster. We used $\gamma = 1.819 \pm 0.003$ [39], and $\nu = 0.875 \pm 0.001$ [40]. These are the exponents for standard 3D percolation. The slope of the dashed line indicates the known value of γ. The averages of this figure were performed over 1000 samples on simple cubic lattices. We used 4 species and $\rho = 0.025$. The axes illustrate how the curves were rescaled. The inset shows the same data in linear scale.

discussed in Sec. V. For the case of $k = 4$, the least-squares fit of the data to a linear function yields $\rho_{max} = 0.159 \pm 0.001$. Note that for $k = 2$ and $k = 3$, p_c never reaches unity for any possible value of initial density ρ, consistent with the polychromatic model.

IV. TOPOLOGICAL PROPERTIES

In this section, we will show that the fractal dimension d_f of the largest cluster at the critical percolation threshold p_c, as well as the fractal dimension of its shortest path, coincide with the established values for three-dimensional (3D) percolation. Additionally, in a regime where several percolating clusters coexist, a phenomenon not previously documented in this context, we analyze interfacial contacts between clusters. We find that the number of such contacts exhibits power-law scaling.

A. Fractal dimension

Next, we study the fractal dimension of these percolating clusters segregated by percolation order i. In fact, we consider for each species the size of the largest cluster at the extrapolated critical threshold $p_c = 0.321$ [46, 47]. Then we define as M_i the mass of the cluster of the species which percolates at the ith threshold $p_c^{(i)}$, averaged over 1000 realizations. The results presented in Fig. 8 suggest that, at the critical point p_c, the least-squares fit to the cluster mass M_i follows [48],

$$M_i = b_i L^{d_f}, \tag{6}$$

where, within error bars, the fractal dimension d_f seems to be always the same and consistent with the fractal dimension of the spanning cluster of classical percolation in 3D ($d_f = 2.5227$ [42, 43]). Additionally, the prefactors b_i, as shown in the inset of Fig. 8, decrease with percolation order i following an exponential law,

$$b_i = Ae^{-Bi}, \tag{7}$$

where the least-squares fit to the data using this exponential law gives $A = 1.02 \pm 0.08$ and $B = 0.29 \pm 0.03$, showing that the average masses of the largest clusters at p_c decrease exponentially with percolation order i.

B. Interfacial contact between the largest clusters at the critical point

Figure 9(a) shows the number of occupied bonds between the largest clusters n_{pc} at p_c. For each contact between two species i and j, n_{pc} follows a power law,

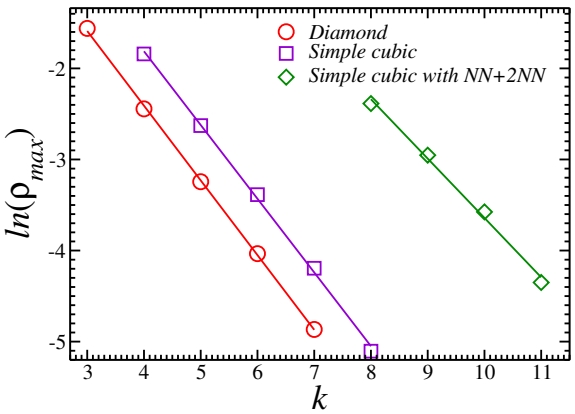

FIG. 15. For each number of species k there is a maximum initial density ρ_{max} above which, in the thermodynamic limit, no percolating cluster exists. The different symbols indicate the results from different 3D lattices. Data are obtained in the following way: for each number of species k, we search for ρ_{max} at which it is possible to perform a finite-size scaling as presented in Fig. 12. We observe that ρ_{max} exhibits an exponential dependence on k, as described by Eq. (12), albeit with different parameters for different lattices as shown in Table I. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

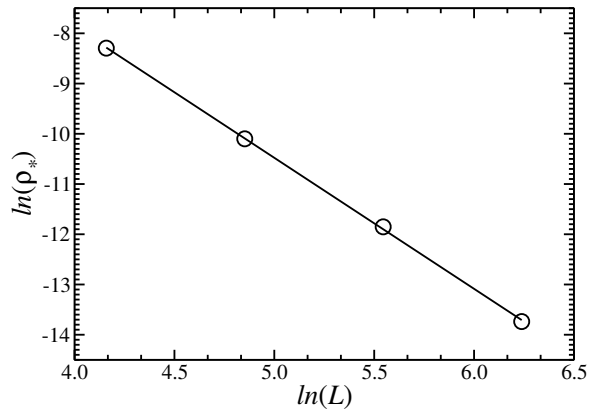

FIG. 16. Logarithmic plot of ρ_* as a function of L. The least-squares fit of the data to a power law, $\rho_*(L) \propto L^{-d_f}$, gives the exponent $d_f = 2.56 \pm 0.04$. We averaged over 1000 realizations for $L = 64$, 100 realizations for $L = 128$ and $L = 256$, and 10 realizations for $L = 512$, on simple cubic lattices. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

$n_{pc}^{i\cap j} \propto L^{d_f^{\cap}}$, where, within the error bars, $d_f^{\cap}$ corresponds to

$$d_f^{\cap} = 2d_f - d, \tag{8}$$

as expected for the intersection between two fractal objects [49].

The same behavior and the corresponding fractal dimensions are seen in Fig. 9(b), where n_p is the number of bonds between the largest clusters at p_c, regardless of whether the bonds had been occupied until p_c, or not.

C. Shortest path

Next, we study the fractal dimension of the shortest path segregated by percolation order i. For each species, we consider the largest cluster at the extrapolated critical threshold $p_c = 0.434$. Then we define l_i the size of the shortest path between the top and bottom of the lattice along the z axis which the cluster percolates at the ith threshold $p_c^{(i)}$, averaged over 10000 realizations. Note that, at p_c, not all species have percolated yet, so we selected only those clusters that had percolated. The results presented in Fig. 10 suggest that, at the critical point p_c the least-squares fit to the shortest path l_i follows [44],

$$l \propto L^{d_{min}}, \tag{9}$$

where, within error bars, the fractal dimension d_{min} seems to be always the same and consistent with the fractal dimension of the shortest path of the spanning cluster of classical percolation in 3D ($d_{min} = 1.3756 \pm 0.0006$ [45]).

We also found the fractal dimension of the shortest path employing the yardstick method [50, 51]. To perform this analysis, we used a cubic lattice with $L = 512$. In this method, we count how many spheres, $N(R)$, of radius R, are needed to cover the entire shortest path. To carry out this count, we first choose a radius R, then select the initial node of the shortest path as the center of the first sphere and identify the point that lies at a distance of $2R$ from the center of the first sphere. This point is used as the center of the next sphere, and the process is repeated successively until the entire shortest path is covered. By repeating the procedure for various values of R, we obtain the corresponding $N(R)$ and derive the well-known relationship:

$$N(R) \propto R^{-d_{min}}. \tag{10}$$

The inset of Fig. 10 shows that the exponents are, within the given precision, the same as the one for standard percolation in 3D, which is $d_{min} = 1.3756 \pm 0.0006$ [45].

V. LIMITS OF THE INITIAL DENSITY

In this section, we explore the conditions required to achieve the maximum number of percolating clusters. Figure 11 shows that the simultaneous percolation of multiple species depends on the system size L, the

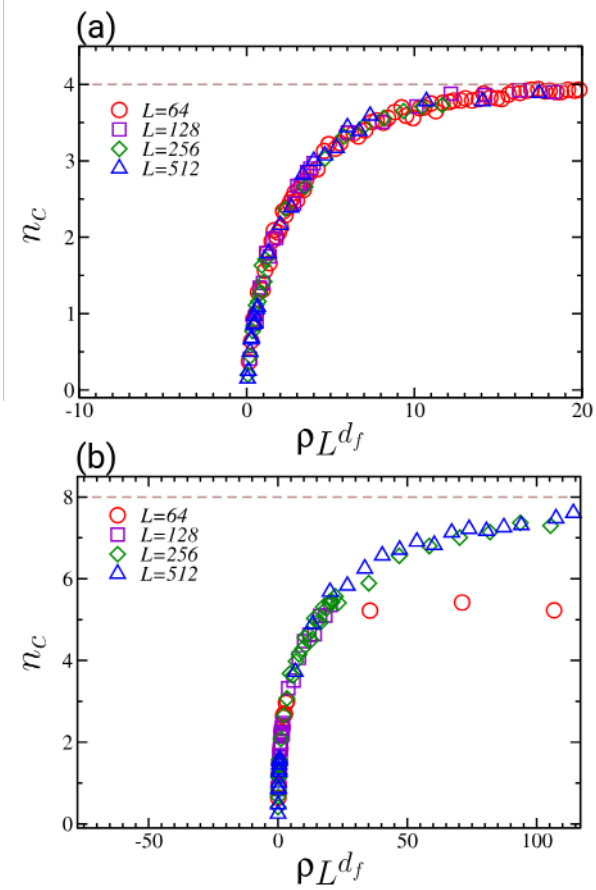

FIG. 17. Average number of percolating clusters for (a) 4 and (b) 8 species, plotted against the rescaled initial density ρ around the region of ρ_{min}. The legend indicates the side length of each simple cubic lattice, and the horizontal axis specifies the rescaling using the fractal dimension for standard 3D percolation ($d_f = 2.5227$ [42, 43]).

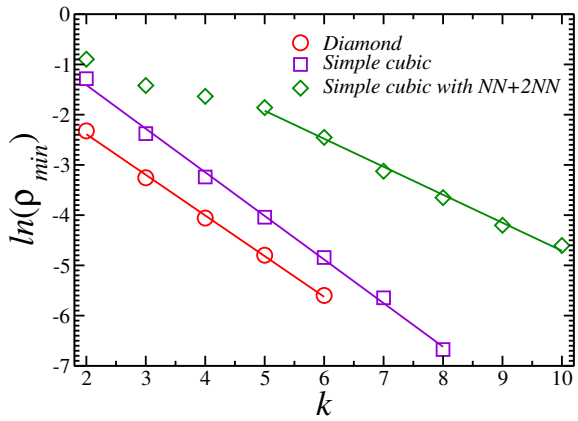

FIG. 18. Semi-logarithmic plot of the minimum density ρ_{min} for obtaining k percolating clusters having the minimum side length L_{min} as a function of the number of species k. The different symbols indicate results for different 3D lattices. Calculating L_{min}, we also obtain ρ_{min}, with the details provided in Sec. VI. We observe that all curves exhibit an exponential law, as described by Eq. (13), albeit with different parameters as shown in Table II, that depend on the lattice structure. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

number of species k, and, also strongly on the initial density ρ. For a given L and k, the initial density ρ must fall within a specific range, $\rho_{min} < \rho < \rho_{max}$, to achieve the maximum number of k percolating clusters.

At the maximum initial density, denoted as ρ_{max}, it appears a transition analogous to the percolation threshold at p_c.

Conversely, the minimum initial density $\rho_{min}(L)$ is size-dependent and vanishes as $L \to \infty$. This implies that in the thermodynamic limit, any nonzero initial density is sufficient to ensure percolation for all species.

A. Maximum initial density ρ_{max}

Since the threshold for site percolation on a simple cubic lattice is $p_c = 0.31160768 \pm 0.00000015 < 1/3$ [42], one obtains simultaneous percolating clusters for $k = 2$ and $k = 3$ even if all sites are inhabited from the beginning ($\rho k = 1$), as in the polychromatic model [24]. On the other hand, if $k > 3$ there exists a $\rho_{max} < 1/k$ above which clusters hinder each other from growing because there are too few uninhabited nodes for the growing clusters to aggregate. In Fig. 12, we see that at this density ρ_{max}, there is a transition from a situation where all species percolate to a situation at which none of the species percolates. As the system size L increases, this transition in ρ becomes more pronounced following a finite-size scaling in $L^{-1/\nu}$, with $\nu = 0.875 \pm 0.001$ being the exponent of standard 3D percolation [40], suggesting that it is a classical percolation transition.

The probability $W(\rho)$ of having n_c percolating clusters in the thermodynamic limit is a step function $\Theta(\rho)$ as follows,

$$\Theta(\rho) = \begin{cases} 1, & \text{if} \quad \rho < \rho_{max} \\ 0, & \text{if} \quad \rho > \rho_{max}. \end{cases} \tag{11}$$

Furthermore, we can analyze the size of the largest cluster of each species S and the second moment χ as a function of ρ, similar to the analysis in Sec. III. We consider the values of S and χ at $p = 1$. In Fig. 13, the curves are labeled by the order of the size of the largest cluster, and we observe a similar behavior to that shown in Fig. 3.

Figure 14 shows that by performing a finite-size scaling of the second moment, one obtains a convincing collapse with the critical exponents γ and ν of standard 3D percolation.

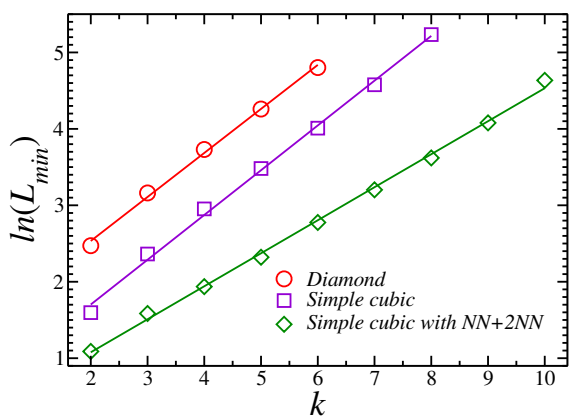

FIG. 19. Semi-logarithmic plot of the minimum lattice side length L_{min} required to obtain k percolating clusters of different species for different types of three-dimensional lattices represented by different symbols. Data are obtained in the following way: for each number of species k, we search for the minimum side length L_{min} for which every species percolates across all realizations. The minimum size L_{min} follows the exponential law with k of Eq. (14). The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

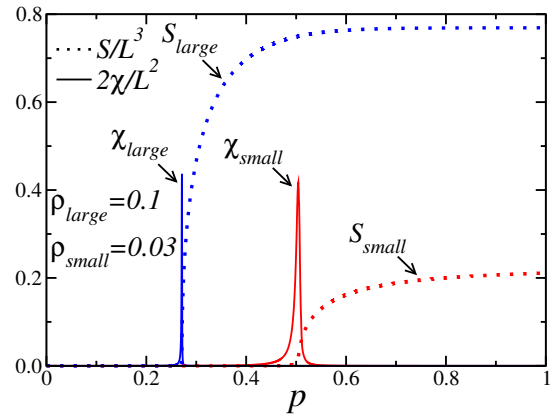

FIG. 20. Evolution of the properties of the cluster distribution during the occupation process for two different initial densities, $\rho_{large} = 0.1$ and $\rho_{small} = 0.03$. This figure presents results similar to those shown in Fig. 3, with different colors representing the results for different species. As observed, the critical points for ρ_{small} and ρ_{large} differ, with the species having the higher initial density percolating first and maintaining a larger mass until the end of the process. Averaging was done over 1000 samples of a simple cubic lattice of size $L = 512$. For convenience each of the quantities was rescaled by the system size, as indicated in the legend.

The density ρ_{max} decreases exponentially with k as

$$\rho_{max} = A_{max} e^{-\alpha_{max} k}, \tag{12}$$

and the least-squares fit to the data using this exponential law for a simple cubic lattice, diamond lattice [37], a simple cubic lattice with bonds between nearest and second nearest neighbors (2NN) [38] yields the parameters listed in Table I, and the results are illustrated in Fig. 15.

Although obtained independently, the exponents α and α_{max} are numerically equal within the margin of error. In fact, using Eqs. (4), (5), and (12), and given that $p_c(\rho_{\max}) = 1$, we obtain the expression

$$e^{(\alpha-\alpha_{max})k} = \frac{1 - C_0^k}{C A_{max}},$$

which implies that if $\alpha = \alpha_{max}$, then C_0^k is independent of k. However, we do not assert this definitively, as our error bars of C_0^k are large.

Table I. Parameters of Eq. (12) for different 3D lattices.

Lattice	Diamond	Simple Cubic	Cubic NN+2NN
A_{max}	2.39 ± 0.04	4.16 ± 0.14	17.7 ± 0.9
α_{max}	0.82 ± 0.01	0.80 ± 0.02	0.65 ± 0.04

B. Minimum density

Let us consider the case $k = 1$. When the initial density ρ is below a certain density ρ_*, uninhabited clusters might grow very much without connecting to any inhabited node and eventually percolate before ever been inhabited. We define ρ_* as the lowest initial density at which the percolation cluster is inhabited at p_c. This minimal density decreases with the system size L as shown in Fig. 16. Indeed, the quantity ρ_* can be interpreted as the probability that an inhabited node belongs to the percolating cluster. Consequently, this probability should be proportional to the fraction of nodes in the percolating cluster, leading to the relation $\rho_* \propto L^{-d_f}$. The least-squares fit to the data of a power law, $\rho_* \propto L^{-d_f}$, gives the exponent $d_f = 2.56 \pm 0.04$, consistent with the fractal dimension of standard 3D percolation.

When the initial density ρ is below ρ_{min}, we observe that at least one species never percolates. The minimum initial density ρ_{min} to obtain the maximum number of k percolating clusters of different species depends on the side length L, as shown in Fig. 17. In the inset of Fig. 11, we observe that the minimum density ρ_{min} depends on L and Fig. 17 shows that in the region around ρ_{min}, the values of n_c for different sizes L collapse on top of each other when the data is rescaled using the exponent

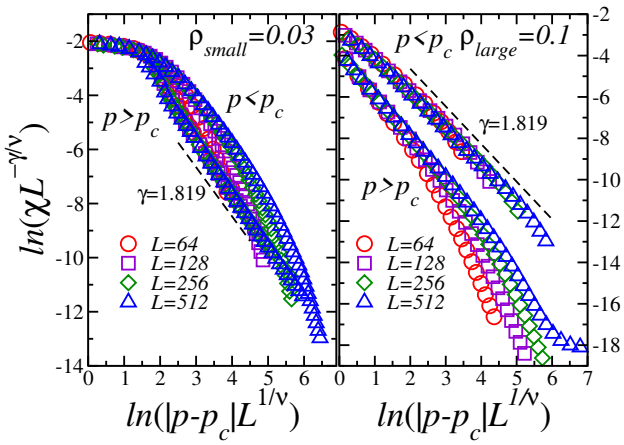

FIG. 21. Logarithmic plot of the finite-size scaling of χ for two different initial densities, $\rho_{small} = 0.03$ (left) and $\rho_{large} = 0.1$ (right) rescaled with $\gamma = 1.819$ and $\nu = 0.875$. The dashed line has slope γ. We observe that despite the difference in the critical points $p_c^{small} = 0.5033 \pm 0.0001$ and $p_c^{large} = 0.2712 \pm 0.0001$, the critical exponents for standard 3D percolation render a good collapse of all the curves. We averaged over 1000 realizations for all sizes L on simple cubic lattices.

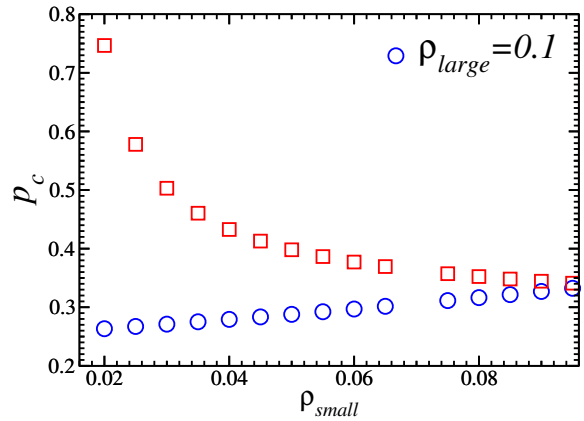

FIG. 22. Percolation thresholds for two different initial densities, ρ_{small} (squares) and $\rho_{large} = 0.1$ (circles). The percolation threshold is the average over 100 realizations. These results were obtained on a cubic lattice of size $L = 512$. We observe that the thresholds come closer as ρ_{small} increases and are equal when the initial densities are the same. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

Table II. Parameters of Eq. (13) for different 3D lattices.

Lattice	Diamond	Simple Cubic	Cubic NN+2NN
A_{min}	0.46 ± 0.08	1.4 ± 0.1	2.37 ± 0.16
α_{min}	0.81 ± 0.02	0.87 ± 0.02	0.56 ± 0.02

$d_f = 2.5227$ for densities $\rho < \rho_{min}$.

VI. THE MINIMUM SIDE LENGTH REQUIRED TO PERCOLATE k CLUSTERS

The density ρ_{min} for the minimum side length L_{min} required to percolate k different species decreases exponentially with k according to the equation:

$$\rho_{min} = A_{min} e^{-\alpha_{min} k}. \tag{13}$$

The least-squares fit to the data using this exponential law for a simple cubic lattice, diamond lattice [37], and a simple cubic lattice with bonds between nearest and second nearest neighbors (2NN) [38] yields the parameters listed in Table II, and the results are illustrated in Fig. 18.

Next, we looked for the minimum lattice size L_{min} required to obtain k percolating clusters of different species considering the cubic lattice with bonds only between the nearest neighbors and the cubic lattice with bonds between the nearest the second nearest neighbors (2NN) [38], as well as the diamond lattice [37]. These results are illustrated in Fig. 19. Each point of this plot is obtained by finding the minimum system size L_{min} for which there are k percolating clusters of different species in all realizations. The curves follow exponential laws,

$$L_{min} = A_L e^{\alpha_L k}. \tag{14}$$

The least-squares fit to the data using Eq. (14) yields the parameters listed in Table III for three different three-dimensional lattices.

We see that the larger the desired number of percolating clusters, the larger the size of the system should be. In other words, we can achieve percolation for as many clusters as we want, as long as we have a lattice large enough.

Table III. Parameters of Eq. 14 for different 3D lattices.

Lattice	Diamond	Simple Cubic	Cubic NN+2NN
A_L	3.98 ± 0.08	1.70 ± 0.01	1.24 ± 0.05
α_L	0.58 ± 0.02	0.59 ± 0.01	0.43 ± 0.01

Moreover, Eqs.(14) and(13) yield a direct relation between $\rho_{\rm min}$ and $L_{\rm min}$, expressed by,

$$\rho_{min} = A_{min} \left(\frac{L_{min}}{A_L} \right)^{-\frac{\alpha_{min}}{\alpha_L}}. \tag{15}$$

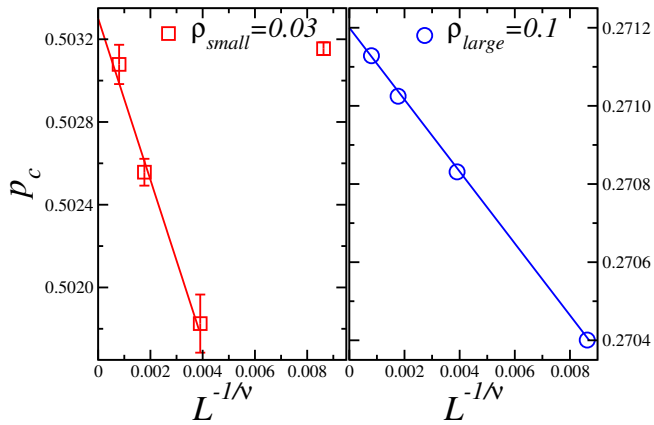

FIG. 23. Percolation threshold p_c as a function of $L^{-1/\nu}$ for two different initial densities, $\rho_{small} = 0.03$ (left) and $\rho_{large} = 0.1$ (right). All realizations have two percolating clusters and the curves follow Eq. (3), giving in the thermodynamic limit, $L \to \infty$, $p_c^{small} = 0.5033 \pm 0.0001$ and $p_c^{large} = 0.2712 \pm 0.0001$. We averaged over 10000 realizations for $L = 64$ and over 1000 for the other sizes L on simple cubic lattices, and used the critical exponent $\nu = 0.875$.

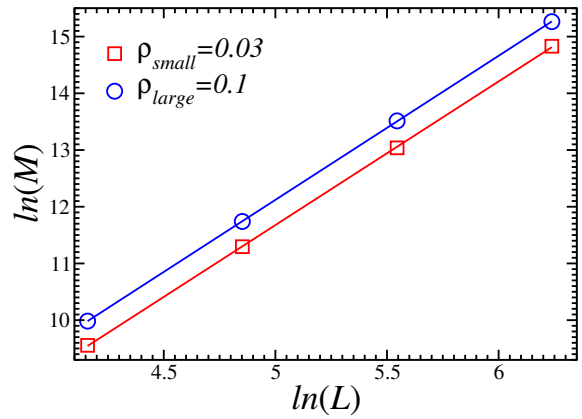

FIG. 24. Fractal dimension for two species with different initial densities $\rho_{small} = 0.03$ and $\rho_{large} = 0.01$. For these values, all realizations have two percolating clusters of different species. The fractal dimensions are obtained through the least-squares fit of the size of the largest cluster of each species at p_c to a power law, $M_i = b_i L^{d_f}$, yielding the exponent $d_f^{small} = 2.53 \pm 0.02$ and $d_f^{large} = 2.54 \pm 0.02$ for the species with density $\rho_{small} = 0.03$ and $\rho_{large} = 0.1$ respectively. Within error bars, our results show that the fractal dimension of the percolating clusters remains the same, regardless of the different initial densities and the observed exponents are consistent with the fractal dimension of standard percolation for 3D. We averaged over 1000 realizations for all sizes L on simple cubic lattices. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

VII. TWO SPECIES WITH DIFFERENT INITIAL DENSITIES

Up to this point, we have exclusively analyzed cases in which all species have the same initial density ρ. In this section, however, we turn our attention to a system with two species, specifically examining the scenario where the initial densities satisfy $\rho_{large} > \rho_{small}$.

A. Evolution of the cluster properties

Figure 20 shows the evolution of the largest cluster S and the second moment χ, as described in Sec. III, for two species with different initial densities, $\rho_{large} = 0.1$ and $\rho_{small} = 0.03$. We observe that the species with the higher initial density, ρ_{large}, has a lower percolation threshold, marked by the maximum of χ, and the largest cluster is larger from the moment it assumes a non-zero value for the species with larger initial density.

Figure 21 shows that, despite the difference in the percolation thresholds, the curves of the second moment χ for different L collapse on top of each other when the data is rescaled using the critical exponents $\gamma = 1.819$ and $\nu = 0.875$ of standard 3D percolation.

B. Percolation threshold

Figure 23 presents the average percolation thresholds p_c^{small} and p_c^{large} for the case of two different densities ρ_{small} and ρ_{large} at which the species exhibits for the first time a cluster that reaches both the bottom and top faces of the system. In contrast to the results of Sec. III B, in the case of two species with different densities, the percolation thresholds do not converge to the same value as L increases. However, the least-squares fit to the data is consistent with Eq. (3), and extrapolating L to $L \to \infty$ gives $p_c^{small} = 0.5033 \pm 0.0001$ and $p_c^{large} = 0.2712 \pm 0.0001$.

The percolation thresholds for both species depend on their initial densities. In Fig. 22, we present the percolation thresholds for both species with $\rho_{large} = 0.1$ fixed and different values of ρ_{small}. The plot shows that the percolation thresholds converge to the same value as ρ_{small} approaches ρ_{large}.

C. Fractal dimension

Next we study the fractal dimension of the percolating clusters. As in Sec. IV A, we consider, for each species, the size of the largest cluster at the extrapolated critical points $p_c^{small} = 0.5033$ and $p_c^{large} = 0.2712$ [46, 47]. Then, we define M as the mass of the largest cluster

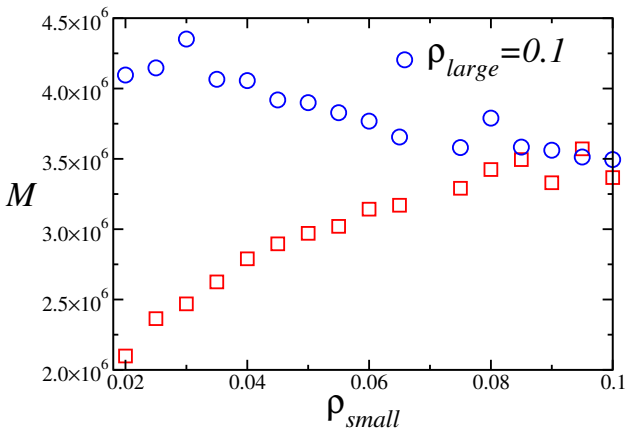

FIG. 25. Mass of the largest cluster at the spanning for two different initial densities, ρ_{small} (squares) and $\rho_{large} = 0.1$ (circles). The mass of the largest is the average over 1000 realizations of the largest cluster when for the first time a cluster connects the bottom and the top face of a simple cubic lattice of size $L = 512$. We observe that the masses of the largest clusters come closer as ρ_{small} increases and are equal when the initial densities are the same. The statistical errors are smaller than the size of the symbols and were therefore omitted from the figure.

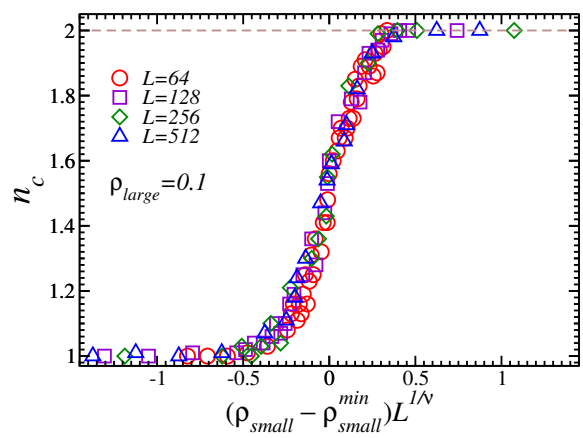

FIG. 26. Finite-size scaling of the average number of percolating clusters of different species, n_c, for two species with different initial densities: ρ_{large} and ρ_{small}. $\rho_{large} = 0.1$ is fixed, and we perform 100 realizations for different densities ρ_{small}, recording each time the number of different percolating clusters n_c obtained. We see that, for a given ρ_{large}, there is a minimum density, ρ_{small}^{min}, above which one finds two percolating clusters of different species. The horizontal axis displays the rescaled variable, using a scaling exponent $\nu = 0.875 \pm 0.001$ [40] and $\rho_{small}^{min} = 0.071 \pm 0.001$. The symbols on the curves represent the sizes L of the the simple cubic lattice, as detailed in the legend.

of the species with initial densities $\rho_{small} = 0.03$ and $\rho_{large} = 0.1$, averaged over 1000 realizations. Using these initial densities, all realizations exhibit two percolating clusters. The results presented in Fig. 24 show that, at the respective critical points p_c, the least-squares fit to the cluster mass M follows Eq. (6), where, within error bars, the fractal dimension d_f is consistent with the fractal dimension of the spanning cluster of classical percolation in 3D ($d_f = 2.5227$ [42, 43]).

The mass M of the largest cluster at p_c for both species depends on the initial densities. In Fig. 25, we present the percolation thresholds for both species with $\rho_{large} = 0.1$ fixed and different values of ρ_{small}. The plot shows that the percolation thresholds converge to the same value as ρ_{small} approaches ρ_{large}.

D. Influence of initial density on the number of percolating clusters

In order to study how the number of percolating clusters of different species, n_c, changes when there are two species with different densities, we analyzed n_c for a fixed $\rho_{large} = 0.1$ while varying ρ_{small}. We find a transition where, as ρ_{small} increases, the system goes from a state with only one percolating species (ρ_{large}) to one with two percolating species. Additionally, as the system size L increases, this transition becomes more pronounced, following a finite-size scaling in $L^{-1/\nu}$, with $\nu = 0.875 \pm 0.001$ being the exponent for standard 3D percolation [40], suggesting that this is a classical percolation transition occurring at ρ_{small}^{min}. Fig. 26 shows

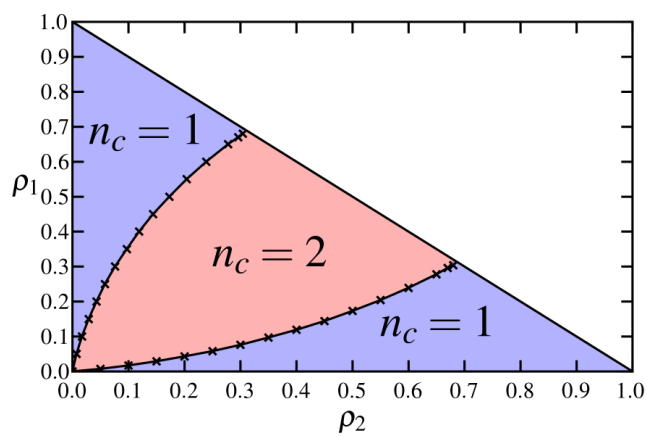

FIG. 27. Phase diagram of the number of percolating clusters for two species with different initial densities ρ_1 and ρ_2. In the red region, two percolating clusters coexist, while in the blue regions, only one cluster percolates. The species that percolates in the blue region is the one with the higher initial density. The points at the intersection between the blue and red regions were obtained through the finite-size scaling shown in Fig. 26. The white region is unphysical, because there $\rho_1 + \rho_2 > 1$.

the finite-size scaling with $\rho_{small}^{min} = 0.071 \pm 0.001$.

Figure 27 presents the phase-diagram of the number of percolating clusters n_c as a function of the two different initial densities ρ_1 and ρ_2. The densities ρ_1 and ρ_2 within the red region correspond to the existence of two percolating clusters $n_c = 2$, while in the blue regions, only the species with higher density percolates, so $n_c = 1$. The white region is unphysical, as it would correspond to $\rho_1 + \rho_2 > 1$.

VIII. CONCLUSIONS

In this work, we have extended the results of Ref. [36], presenting a model that allows for the existence of an unlimited number of percolating clusters both at and above the critical point. We found that the minimum lattice size L_{min} required to achieve a specific number k of percolating clusters grows exponentially with k. In other words, if we have a sufficiently large lattice, we can obtain as many percolating clusters as desired. The maximum initial density ρ_{max} decreases exponentially with k. The various percolation transitions all seem to have the critical exponents of 3D classical percolation and thus to belong to the same universality class. Furthermore, for the case $k = 2$, when ρ is not the same for both species, fixing one of densities ρ_{large}, there is a minimum density ρ_{small}^{min} above which there is a transition from where only one species percolates to where both percolate. Remains as challenge for the future to explain the origin of Eq. (14) namely the logarithmic dependence of the maximum number of possible spanning clusters on system size. The findings of our study strongly suggest that it should be possible to experimentally realize objects exhibiting multiple percolating clusters such as trigels, tetragels, and so on.

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